BRAINWARE UNIVERSITY
SCHOOL OF COMPUTATIONAL & APPLIED SCIENCES
DEPARTMENT OF COMPUTATIONAL SCIENCES
BACHELOR OF COMPUTER APPLICATIONS /
BACHELOR OF COMPUTER APPLICATIONS (HONOURS) /
BACHELOR OF COMPUTER APPLICATIONS (HONOURS with RESEARCH) – 2024
SEMESTER – IV

Sl. No.	Course Code	Course	L	T	P	Evaluation Scheme		Total	Credits	Course Type
						CIA	ESE			
1	BCA40107	Object Oriented Programming in JAVA	3	0	0	40	60	100	3	Major
2	BCA49108	Object Oriented Programming in JAVA Lab	0	0	4	40	60	100	2	Major
3	BCA49109	PHP and MySQL Lab	0	0	4	40	60	100	2	Major
4	BCA40110	Operating System	3	0	0	40	60	100	3	Major
5	BCA47111	Full-stack Development-I	2	0	4	40	60	100	4	Major
6	BCA40204	Data Warehousing and Data Mining	4	0	0	40	60	100	4	Minor
7	AEC09001	Communicative English Lab	0	0	4	40	60	100	2	AEC
TOTAL			**12**	**0**	**16**	**280**	**420**	**700**	**20**	

BRAINWARE UNIVERSITY
SCHOOL OF COMPUTATIONAL & APPLIED SCIENCES
DEPARTMENT OF COMPUTATIONAL SCIENCES
BACHELOR OF COMPUTER APPLICATIONS /
BACHELOR OF COMPUTER APPLICATIONS (HONOURS) /
BACHELOR OF COMPUTER APPLICATIONS (HONOURS with RESEARCH) – 2024
SEMESTER – IV

Course Code: BCA40107
Course Name: Object Oriented Programming in JAVA
Weekly Contact Hours: 3L
Credits: 3
Total Allotted Hours: 45L

Course Objective: The aim of this course is to provide an introduction to object oriented programming (OOP) using the Java programming language. Its main objective is to teach the basic concepts and techniques which form the object oriented programming paradigm.

Pre-requisite(s): Knowledge of programming language.

Course Outcomes (COs): After the completion of the course, students would be able to:

CO1: Define the concept of object oriented design and compare OOPs with other programming languages.
CO2: Illustrate basic structure of Java program and apply different Java methods, constructs and libraries for real world applications.
CO3: Apply and analyze inheritances, interfaces and packages of Java.
CO4: Analyze and assess exception handling of Java.
CO5: Distinguish and develop threads and applet.

Module 1: Object Oriented Design **[4H]**

Concepts of object-oriented programming language, object, class, relationships among objects, relationships among classes-association, aggregation using instantiation, meta-class, difference between OOP and procedural language – advantages and disadvantages, concepts of message passing, overview of the properties of OOPs.

Module 2: Class and Object Properties **[16H]**

Basic concepts of java programming – advantages of java, byte-code & JVM, data types, access specifiers, operators, control statements and loops, array, creation of class, object, constructor, finalize and garbage collection, use of method overloading, this keyword, use of objects as parameter and methods returning objects, call by value and call by reference, static variables and methods, nested and inner classes, basic string handling concepts- string functions, concept of mutable and immutable string, command line arguments, basics of I/O operations – keyboard input using Buffered Reader and Scanner classes.

Module 3: Inheritance and Packages **[9H]**

Definition, types of inheritance, super class and subclasses including multilevel hierarchy, use of super and final keywords, dynamic method dispatch, use of abstract classes and methods, interfaces. creation of packages, importing packages, member access for packages.

BRAINWARE UNIVERSITY
SCHOOL OF COMPUTATIONAL & APPLIED SCIENCES
DEPARTMENT OF COMPUTATIONAL SCIENCES
BACHELOR OF COMPUTER APPLICATIONS /
BACHELOR OF COMPUTER APPLICATIONS (HONOURS) /
BACHELOR OF COMPUTER APPLICATIONS (HONOURS with RESEARCH) – 2024
SEMESTER – IV

Module 4: Exception Handling **[6H]**

Exception handling basics, different types of exception classes, use of try and catch with throw, throws and finally, creation of user defined exception classes.

Module 5: Threads & Applet **[10H]**

Basics of multithreading, main thread, thread life cycle, creation of multiple threads, thread priorities, thread synchronization, inter-thread communication, deadlocks for threads, suspending and resuming threads, basics of applet programming, applet life cycle, event model and listener, creation of buttons & text fields.

Course Outcomes (COs) and Modules Mapping:

Sl. No	Course outcome (COs)	Mapped Module(s)	Learning levels (as per Bloom's Taxonomy from K1 to K6)
1.	CO1	M1	K1, K2
2.	CO2	M2	K2, K3
3.	CO3	M3	K3, K4
4.	CO4	M4	K4, K5
5.	CO5	M5	K5, K6

Text Books:

1. Programming with Java: A Primer, E. Balagurusamy, TMH, 4th Edition, 2006.

Reference Books:

1. The complete reference-Java2, Patrick Naughton, Herbert Schildt, TMH, 5th Edition, 2017
2. Object Oriented System Development, Ali Bahrami, McGraw Hill, Classic Edition, 2017.

BRAINWARE UNIVERSITY
SCHOOL OF COMPUTATIONAL & APPLIED SCIENCES
DEPARTMENT OF COMPUTATIONAL SCIENCES
BACHELOR OF COMPUTER APPLICATIONS /
BACHELOR OF COMPUTER APPLICATIONS (HONOURS) /
BACHELOR OF COMPUTER APPLICATIONS (HONOURS with RESEARCH) – 2024
SEMESTER – IV

Course Code: BCA49108
Course Name: Object Oriented Programming in JAVA Lab
Weekly Contact Hours: 4P
Credits: 2
Total Allotted Hours: 60P

Course Objective: The aim of this course is to provide an introduction to object oriented programming (OOP) using the Java programming language. Its main objective is to teach the basic concepts and techniques which form the object oriented programming paradigm.

Pre-requisite(s): Knowledge of programming language.

Course Outcomes (COs): After the completion of the course, students would be able to:

CO1: Write and analyze programs related to the concept of OOPs.
CO2: Implement classes and create objects with emphasis on data hiding.
CO3: Experiment and analyze polymorphism and inheritances in Java.
CO4: Test and assess exception handling and multithreading of Java.
CO5: Simulate applet programming and compare different Collections.

Module 1: Object Oriented Design **[8H]**

Basic Java programs with different data types, branching and iterative statements, compilation and interpretation of Java programs.

Module 2: Class & Object Properties **[8H]**

Classes and objects, implementation of access control modifiers in class

Module 3: Polymorphism and Inheritance **[20H]**

Programs with command line arguments, array and string, function overloading and overriding, constructor, constructor overloading, static data members and methods, inheritance by extending classes, inheritance with interfaces, packages

Module 4: Exception Handling & Multithreading **[12H]**

Thread using Thread class and Runnable interface, try, catch, throw and throws examples, user defined exception handling.

Module 5: Applet Programming and advanced topics **[12H]**

Array List, Vector, List, Map, HashMap, Set, HashSet and their detailed operations.

BRAINWARE UNIVERSITY
SCHOOL OF COMPUTATIONAL & APPLIED SCIENCES
DEPARTMENT OF COMPUTATIONAL SCIENCES
BACHELOR OF COMPUTER APPLICATIONS /
BACHELOR OF COMPUTER APPLICATIONS (HONOURS) /
BACHELOR OF COMPUTER APPLICATIONS (HONOURS with RESEARCH) – 2024
SEMESTER – IV

Course Outcomes (COs) and Modules Mapping:

Sl. No.	Course Outcome (COs)	Mapped Module(s)	Learning levels (as per Bloom's Taxonomy from K1 to K6)
1.	CO1	M1	K4, K5
2.	CO2	M2	K5, K6
3.	CO3	M3	K4
4.	CO4	M4	K4, K5
5.	CO5	M5	K4. K5, K6

Text Books:

1. Programming with Java: A Primer, E. Balagurusamy, TMH, 3rd Edition, 2006.

BRAINWARE UNIVERSITY
SCHOOL OF COMPUTATIONAL & APPLIED SCIENCES
DEPARTMENT OF COMPUTATIONAL SCIENCES
BACHELOR OF COMPUTER APPLICATIONS /
BACHELOR OF COMPUTER APPLICATIONS (HONOURS) /
BACHELOR OF COMPUTER APPLICATIONS (HONOURS with RESEARCH) – 2024
SEMESTER – IV

Course Code: BCA49109
Course Name: PHP and MySQL Lab
Weekly Contact Hours: 4P
Credits: 2
Total Allotted Hours: 60P

Course Objective: This course aims to provide a comprehensive grasp of PHP programming, covering syntax, variables, loops, HTML integration, string manipulation, functions, arrays, complex database operations and dynamic web applications, focusing on user interactions and database functionalities.

Pre-requisite(s): Knowledge in basic programming.

Course Outcomes (COs): After completion of this course, students will be able to:

CO1: Develop and evaluate basic PHP programs integrated with HTML.
CO2: Analyze and evaluate PHP functions and string operations, focusing on recursion, string manipulation, and the use of library functions.
CO3: Plan and evaluate PHP array management techniques, encompassing both index-based and associative arrays with control statements.
CO4: Analyze and simulate PHP techniques for managing HTML forms, including data processing, file operations, and form redirection.
CO5: Develop and evaluate a PHP-MySQL project demonstrating database connectivity, basic SQL operations, and exception handling.

Module 1: Introduction to PHP **[12H]**

Demonstrate simple task (such as: to display 'hello') and basic PHP Syntax (includes defining variable, constant, data type and basic arithmetic operations with operators). demonstrate condition/decision making and loop statement. Mixing of decision and looping statements with HTML script.

Module 2: Function and String in PHP **[8H]**

Call by value, call by reference and recursive function. String creating and accessing, string searching & replacing string, formatting string and demonstrating string related library function.

Module 3: Array in PHP **[12H]**

Creating index based and associative array. Also demonstrate accessing the array elements using loop, associative array using for each () loop and few related library function. Array by including decision and loop statement.

BRAINWARE UNIVERSITY
SCHOOL OF COMPUTATIONAL & APPLIED SCIENCES
DEPARTMENT OF COMPUTATIONAL SCIENCES
BACHELOR OF COMPUTER APPLICATIONS /
BACHELOR OF COMPUTER APPLICATIONS (HONOURS) /
BACHELOR OF COMPUTER APPLICATIONS (HONOURS with RESEARCH) – 2024
SEMESTER – IV

Module 4: Handling Html Form with PHP **[12H]**

Demonstrate to capture user inputs through form. Generating file uploaded form and redirecting a form after submission, Handling file & directory related operations (opening, closing, copying, renaming and deleting a file and directory).

Module 5: PHP MySQL Connectivity and Exception handling **[16H]**

Performing database connectivity and other basic database oriented operations. A mini project based learning of PHP MySQL database connectivity with basic database operation (DML, like Insert, delete, update, Select) and setting query parameter, executing query- join with capability of exception and error handling.

Course Outcomes (COs) and Modules Mapping:

Sl. No.	Course Outcome (COs)	Mapped Module(s)	Learning levels (as per Bloom's Taxonomy from K1 to K6)
1.	CO1	M1	K6,K5
2.	CO2	M2	K4,K5
3.	CO3	M3	K4,K5
4.	CO4	M4	K4,K6
5.	CO5	M5	K5, K6

Text books:

1. PHP and MySQL Web Development, Luke Welling, Laura Thomson, Pearson,5th Edition.

Reference Books:

1. Programming PHP: Creating Dynamic Web Pages, Tatroe K., MacIntyre P., & Lerdorf, R., O'Reilly Press 2013, 3rd edition.

BRAINWARE UNIVERSITY
SCHOOL OF COMPUTATIONAL & APPLIED SCIENCES
DEPARTMENT OF COMPUTATIONAL SCIENCES
BACHELOR OF COMPUTER APPLICATIONS /
BACHELOR OF COMPUTER APPLICATIONS (HONOURS) /
BACHELOR OF COMPUTER APPLICATIONS (HONOURS with RESEARCH) – 2024
SEMESTER – IV

Course Code: BCA40110
Course Name: Operating System
Weekly Contact Hours: 3L
Credits: 3
Total Allotted Hours: 45L

Course Objective: The aim of this course is to provide an in-depth study of operating systems, their design principles, and their role in computer systems. It covers the fundamental concepts, algorithms, and mechanisms used in operating systems.

Pre-requisite(s): Basic knowledge in computer.

Course Outcomes (COs): After the completion of the course, students would be able to:

CO1: Gain a solid understanding of the fundamental concepts and principles underlying operating systems, including process management, memory management, file systems, and scheduling algorithms.
CO2: Apply knowledge of operating system concepts to analyze and solve real-world problems related to process synchronization, memory allocation, resource management, and system performance optimization.
CO3: Develop practical skills by implementing or simulating operating system components such as process schedulers, memory management algorithms, file systems, or device drivers.
CO4: Evaluate and compare different operating system designs and architectures.
CO5: Learn techniques for measuring and evaluating the performance of operating systems, identifying performance bottlenecks, and proposing optimization strategies to improve system efficiency.

Module 1: Basics of Operating System **[4H]**

Overview of operating system functions and goals, history and evolution of operating systems, types of operating systems, user interfaces and system calls.

Module 2: PROCESS MANAGEMENT: **[16H]**
Process concepts, process Attributes, process state, process control block, Swapping ,process Scheduler, process scheduling, process scheduling algorithms.
CONCURRENCY AND SYNCHRONIZATION: Process synchronization, critical section problem, Peterson's solution, and condition variables classical synchronization problems-producer-consumer, readers and writers problem, dining philosophers problem, semaphores.
DEADLOCKS: System model, deadlock characterization, deadlock prevention, detection and avoidance, recovery from deadlock banker's algorithm.

Module 3: Memory Management **[12H]**
Memory hierarchy and organization, memory allocation strategies - contiguous, non-contiguous, paging , structure of the page table and segmentation techniques, virtual memory concepts and address translation, demand paging memory management unit (MMU) and page replacement algorithms.

BRAINWARE UNIVERSITY
SCHOOL OF COMPUTATIONAL & APPLIED SCIENCES
DEPARTMENT OF COMPUTATIONAL SCIENCES
BACHELOR OF COMPUTER APPLICATIONS /
BACHELOR OF COMPUTER APPLICATIONS (HONOURS) /
BACHELOR OF COMPUTER APPLICATIONS (HONOURS with RESEARCH) – 2024
SEMESTER – IV

Module 4: File Systems and Input/output Systems **[8H]**

File system organization and file attributes, file operations, file allocation methods, directory structure and file system implementation, access control and file permissions, I/O devices, controllers, and interfaces, I/O operations and buffering, I/O scheduling algorithms and device drivers, interrupt handling and I/O software layers, disk management and file system I/O.

Module 5: Distributed System **[5H]**

Distributed system concepts and architectures, distributed file systems and distributed process management.

Course Outcomes (COs) and Modules Mapping:

Sl. No	Course Outcome (COs)	Mapped Module(s)	Learning levels (as per Bloom's Taxonomy from K1 to K6)
1.	CO1	M1	K1, K2
2.	CO2	M2	K3, K4,
3.	CO3	M2, M3	K5, K6
4.	CO4	M5	K4, K5
5.	CO5	M2, M4	K2, K4, K5

Text Books:

1. Operating System Concepts, Abraham Silberschatz, Peter B. Galvin, and Greg Gagne, Wiley Publisher.

Reference Books:

1. Operating Systems: Internals and Design Principles, William Stallings, Pearson Education India, 7th Edition, 2013.
2. Modern Operating Systems, Andrew S. Tanenbaum and Herbert Bos. Pearson.

BRAINWARE UNIVERSITY
SCHOOL OF COMPUTATIONAL & APPLIED SCIENCES
DEPARTMENT OF COMPUTATIONAL SCIENCES
BACHELOR OF COMPUTER APPLICATIONS /
BACHELOR OF COMPUTER APPLICATIONS (HONOURS) /
BACHELOR OF COMPUTER APPLICATIONS (HONOURS with RESEARCH) – 2024
SEMESTER – IV

Course Code: BCA47111
Course Name: Full-Stack Development-I
Weekly Contact Hours: 2L+ 4P
Credits: 4
Total Allotted Hours: 30L+ 60P

Course Objective: This course aims to equip students with a comprehensive understanding of Node.js and its practical applications. Students will develop the fundamental concepts of Node.js, including its definition, advantages, and real-world use cases.

Pre-requisite(s): Knowledge of basic web technologies such as HTML, CSS, JavaScript.

Course Outcomes (COs): After completion of this course, students will be able to:

CO1: Understand and evaluate Node.js based web application.
CO2: Apply and assess Node.js module along with NPM and package.json file.
CO3: Create and operate a basic web server with HTTP module and handle requests/responses, and perform essential file system operations.
CO4: Employ advanced debugging with Node.js debugger and develop/compose Visual Studio Code for event-driven programming, and configure routes in Express.js.
CO5: Develop a functional web application and employ Node.js and MongoDB, showcasing proficiency in CRUD operations, connection management, and middleware integration.

Module 1: Introduction to Node-JS and Setup Development Environment **[4H(L)]**
Definition and usage of Node.js, advantages of Node.js, use cases and real-world examples, traditional web server model vs Node.js process model, installing Node.js on Windows / MacOS, setting up a basic development environment.

Practical: **[6H(P)]**
Working in REPL (Node.js interactive shell), Node.js console and basic commands.

Module 2: Node JS Modules and Node Package Manager **[4H(L)]**
Functions in JavaScript, introduction to buffers, Node.js modules, core modules, local modules, require (). Overview of the Node Package Manager (NPM), installing packages locally and globally, managing dependencies with package.json, updating packages.

Practical: **[12H(P)]**
Implementation of functions in JavaScript, Node.js modules, core modules, local modules, require (), installing packages locally and globally, managing dependencies with package.json, updating packages.

Module 3: Creating a Basic Web Server using HTTP Module **[6H(L)]**
Handling HTTP requests and responses, sending requests to the server, Introduction to the file system module (fs), Reading and writing files synchronously and asynchronously, opening, updating, and deleting files, other input/output

operations.

Practical: **[14H(P)]**
Working with handling HTTP requests and responses, sending requests to the server, Introduction to the file system module (fs), reading and writing files synchronously and asynchronously, opening, updating, and deleting files, other input/output operations.

Module 4: Debugging Node.js Application, Events, Express.js **[6H(L)]**
Core Node.js Debugger, debugging with Visual Studio code, Understanding and utilizing EventEmitter class, returning event emitters from functions, inheriting events in Node.js, Configuring routes in Express.js, Building a basic web application with Express.js, middleware in Express.js.

Practical: **[12H(P)]**
Working with Node.js applications. Configuring routes in Express.js and events.

Module 5: Serving Static Resources and Database Connectivity **[10H(L)]**
Serving static files with Express.js, working with middleware in Express.js, NoSQL, Introduction to MongoDB and its use cases, connecting to MongoDB from Node.js, CRUD operations with MongoDB, reading data with the select command, updating records, deleting records, Managing connection strings and configurations.

Practical: **[16H(P)]**
Implementation of serving static files with Express.js, working with middleware in Express.js, NoSQL, connecting to MongoDB from Node.js, CRUD operations with MongoDB, reading data with the select command, updating records, deleting records, Managing connection strings and configurations.

Course Outcomes (COs) and Modules Mapping:

Sl. No.	Course Outcome (COs)	Mapped Module(s)	Learning levels (as per Bloom's Taxonomy from K1 to K6)
1.	CO1	M1	K2, K5
2.	CO2	M2	K3,K5
3.	CO3	M3	K3, K6
4.	CO4	M4	K3,K4
5.	CO5	M5	K3, K6

Text books:
1. Node.js, MongoDB and Angular Web Development, Brad Dayley, Brendan Dayley, Caleb Dayley, Addison-Wesley,2nd Edition.

Reference books:
1. Full Stack Development with MongoDB, Sharma, Manu. BPB Publications, 2022.
2. Node.js in Action, Alex Young, Bradley Meck, and Mike Cantelon, Manning,2nd Edition
3. Node.js by Example, Tsonev, Packt Pub Ltd.

BRAINWARE UNIVERSITY
SCHOOL OF COMPUTATIONAL & APPLIED SCIENCES
DEPARTMENT OF COMPUTATIONAL SCIENCES
BACHELOR OF COMPUTER APPLICATIONS /
BACHELOR OF COMPUTER APPLICATIONS (HONOURS) /
BACHELOR OF COMPUTER APPLICATIONS (HONOURS with RESEARCH) – 2024
SEMESTER – IV

Course Code: BCA40204
Course Name: Data Warehousing and Data Mining
Weekly Contact Hours: 4L
Credits: 4
Total Allotted Hours: 60L

Course Objective: The aim of the course is to understand the fundamental concepts and principles of data warehouse and Datamining, including its applications and the data mining process

Pre-requisite(s): Strong understanding of statistics, mathematics, programming, and database systems.

Course Outcomes (COs): After the completion of the course, students would be able to:

CO1: Define and understand the fundamental concepts and principles of data mining, including its applications and the data mining process.
CO2: Explain and apply various types of Datawarehouse architecture and Operations.
CO3: Illustrate and utilize a range of supervised and unsupervised learning algorithms to perform classification, regression, clustering, and association rule mining tasks.
CO4: Evaluate and validate data mining models using Clustering and Association rule techniques, and make informed decisions based on the results.
CO5: Apply advanced data mining techniques to formulate and analyse text data, social networks, time series data, and big data environments.

Module 1: Introduction to Data Mining **[10H]**

Overview of data mining and its applications, data mining process and methodologies, KDD, data pre-processing: data cleaning, integration, transformation, and reduction, Exploratory data analysis and visualization techniques.

Module 2: Data Warehouse **[12H]**

Introduction to Data Warehouse, differences between operational data base systems and data warehouse, data warehouse characteristics, data warehouse architecture and its components, extraction Transformation-Loading, Logical (Multidimensional), data modelling, schema design, star and snow-flake Schema, fact constellation, fact table, OLAP cube, OLAP operations, OLAP server architecture-ROLAP, MOLAP and HOLAP.

Module 3: Classification and Regression **[12H]**

Overview of classification and regression problems in data mining, decision tree algorithms (C4.5, CART), Bayesian classifiers (naive Bayes, Bayesian networks), Evaluation metrics for classification and regression models (accuracy, precision, recall, F1 score, mean squared error), Ensemble methods (bagging, boosting, random forests).

BRAINWARE UNIVERSITY
SCHOOL OF COMPUTATIONAL & APPLIED SCIENCES
DEPARTMENT OF COMPUTATIONAL SCIENCES
BACHELOR OF COMPUTER APPLICATIONS /
BACHELOR OF COMPUTER APPLICATIONS (HONOURS) /
BACHELOR OF COMPUTER APPLICATIONS (HONOURS with RESEARCH) – 2024
SEMESTER – IV

Module 4: Clustering and Association Rule Mining **[13H]**

Clustering algorithms (k-means, hierarchical clustering, DBSCAN), Density-based clustering (OPTICS, Mean Shift), Association rule mining (Apriori algorithm, FP-growth algorithm), evaluation metrics for clustering and association rule mining (silhouette coefficient, lift, confidence, support).

Module 5: Advanced Topics in Data Mining **[13H]**

Text mining and sentiment analysis, Social network analysis and link analysis, time series analysis and forecasting, data mining in big data and distributed environments, ethical considerations and privacy issues in data mining.

Course Outcomes (COs) and Modules Mapping:

Sl. No.	Course Outcome (COs)	Mapped Module(s)	Learning levels (as per Bloom's Taxonomy from K1 to K6)
1.	CO1	M1	K1, K2
2.	CO2	M2,M3,	K2,K3, K5
3.	CO3	M2,M3	K2,K3
4.	CO4	M4	K4, K5
5.	CO5	M5	K3, K4, K6

Text books:

1. Data Mining and Data Warehousing, Parteek Bhatia, Cambridge University Press, 2nd ed., 2021.

Reference books:

1. Data Mining: Concepts and Techniques, Han, J., Kamber, M., & Pei, J., Morgan Kaufmann, 3rd ed., 2012.
2. Data Warehousing, Data Mining & OLAP , Alex Berson and Stephen J.Smith, Tata McGraw – Hill Edition, 35th Reprint, 2016.

BRAINWARE UNIVERSITY
SCHOOL OF COMPUTATIONAL & APPLIED SCIENCES
DEPARTMENT OF COMPUTATIONAL SCIENCES
BACHELOR OF COMPUTER APPLICATIONS /
BACHELOR OF COMPUTER APPLICATIONS (HONOURS) /
BACHELOR OF COMPUTER APPLICATIONS (HONOURS with RESEARCH) – 2024
SEMESTER – IV

Course Code: AEC09001
Course Name: Communicative English Lab
Weekly Contact Hours: 4P
Credits: 2
Total Allotted Hours: 60P

Course Objective: The objective of this course is to build communication skills in students through various listening, speaking, reading and writing activities aided by audio-visuals. The goal is to make the students proficient users of the English language in professional and real-life situations.

Pre-requisite(s): Basic Knowledge in English.

Course Outcomes (COs): After completion of the course, students will be able to:

CO1: Engage in effective listening with the aim of understanding the content, recalling key-points and applying the same in real-life conversations and contexts while speaking.
CO2: Develop their speaking ability through observation of audio-visual content that aids them in individual and group-based activities simulating real-life scenarios.
CO3: Develop various language skills, verbal and non-verbal, and analyze them contextually and apply them effectively to face interviews, engage in group discussions, and public speaking.
CO4: Design, create, and deliver professional PowerPoint Presentations and justify their arguments effectively.

Various Modules of this course will be conducted with suitable activities with the help of Computer Assisted Learning Mode and / or Instructor Led Interactive Learning Mode.

Module 1: Learning the English Language through Effective Listening [12H]

Listening to motivational speeches and success stories of renowned personalities and summarizing the key aspects. Listening for specific information and note-taking. Listening to telephone conversations.

Module 2: Learning the English Language through Effective Speaking [12H]

Learning to introduce yourself. Learning to greet and compliment properly. Learning to criticize and disagree politely. Learning to negotiate skillfully. Learning effective telephonic conversation.

Module 3: Enhancing Speaking Skills through Individual and Group-based Activities [12H]

Just A Minute (JAM) and Extempore Activities. Debate, Group Discussion, Public Speaking--Delivering Welcome Speech, Public Speaking---Delivering Vote of Thanks.

BRAINWARE UNIVERSITY
SCHOOL OF COMPUTATIONAL & APPLIED SCIENCES
DEPARTMENT OF COMPUTATIONAL SCIENCES
BACHELOR OF COMPUTER APPLICATIONS /
BACHELOR OF COMPUTER APPLICATIONS (HONOURS) /
BACHELOR OF COMPUTER APPLICATIONS (HONOURS with RESEARCH) – 2024
SEMESTER – IV

Module 4: Role Play **[12H]**

Conversations between colleagues, Conversations between supervisor and subordinate, Conversations between salesmen and customers, and Everyday conversations (travel booking, banking, shopping.)

Module 5: Introducing PowerPoint Presentation Skills **[12H]**

Verbal presentations: Content delivery through effective expressions, proper pronunciation, voice modulation, right body language, and professional etiquette, Hands-on Individual and Group presentation using Microsoft PowerPoint, understanding your audience, and responding to questions.

Course Outcomes (COs) and Modules Mapping:

Sl. No.	Course Outcome (COs)	Mapped Module(s)	Learning levels (as per Bloom's Taxonomy from K1 to K6)
1.	CO1	M1, M2, M4	K1, K2, K3
2.	CO2	M1, M2	K1, K6
3.	CO3	M2, M3, M4	K3, K4, K6
4.	CO4	M5	K3, K5, K6

Text Books:

1. English Language Laboratories: A Comprehensive Manual, Nira Konar, PHI Publication, 2nd Edition.
2. A Manual for English Language laboratory, D. Sudha Rani, Pearson Publication, 2011 Edition.

Reference Books:

1. Handbook of Standard English and Indian Usage. Sethi J. A. New Delhi India: Prentice- Hall, 2007.